

Md Tanzimul Islam Refat

Machine Learning · Reliable Systems

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SUMMARY

Computing and Information Systems undergraduate at Daffodil International University exploring machine learning for reliable software systems, with research in federated learning and cross-system anomaly detection for microservices. Recently completed a Machine Learning internship at FlyRank AI and serve as Research & Development Secretary at AIRIS, DIU. Also build full-stack web applications and cloud-native deployment pipelines.

EDUCATION

B.Sc. in Computing and Information Systems

Daffodil International University

2023 – 2027 (expected)

CGPA: 3.50 / 4.00 (up to 7th semester)

Higher Secondary Certificate (HSC), Science

Lions School and College, Saidpur, Nilphamari

2021 – 2022

GPA: 5.00 / 5.00

EXPERIENCE

Machine Learning Intern · FlyRank AI (Remote)

Jul 2026 – Sep 2026

- Completed a machine learning internship focused on model development and research.

Research & Development Secretary · AIRIS, Daffodil International University

Feb 2025 – Present

- Lead AI research initiatives and coordinate collaborative student research projects.

Career Ambassador · DIU Career Development Center

May 2025 – May 2026

- Supported campus career events and student engagement.

RESEARCH

FedTransferAD — Federated Cross-System Microservice Anomaly Detection with Semantic Metric Harmonization

Submitted to ICECE 2026, BUET, Dhaka

[project page](#)

- Mapped heterogeneous telemetry into 7 semantic metric families and a shared 21-feature representation.
- Trained a 22-parameter classifier with weighted FedAvg on two systems, tested on an unseen third.
- Mean F1 **0.8905**, ROC-AUC 0.9413, AP 0.9601 on RCAEval RE1 (3 held-out systems); explained with SHAP.

PROJECTS

Cloud-Native GitOps Pipeline

[code](#) · [config](#)

GitHub Actions, Docker, Kubernetes, FluxCD

- Automated build and deployment pipelines for a containerized Node.js application using GitOps.

AURFIA — Jewelry E-commerce Platform

[code](#)

Next.js, PostgreSQL, Drizzle ORM, Stripe, Docker, AWS

- 150+ products, search and filters, Stripe Checkout, and an admin dashboard; deployed with Docker on AWS.

SKILLS

ML & Research: Machine learning, federated learning, anomaly detection, time-series / telemetry analysis

Languages: Python, JavaScript / TypeScript, SQL

Web: Next.js, Node.js, PostgreSQL, Drizzle ORM, Stripe

Cloud & DevOps: Docker, Kubernetes, FluxCD, GitHub Actions, AWS

CERTIFICATES & RECOGNITION

- Machine Learning Internship** — FlyRank AI, Sep 2026 Credential ID: FR-D11-F3243-A3952
- ICPC Online Preliminary Contest** — **Honorable Mention**, 2024
- CS50's Introduction to Programming with Python** — Harvard University, 2026
- Intro to Machine Learning** — Kaggle, Sep 2026
- CDC Career Ambassador** — Daffodil International University, May 2026